

When Clustered Extremes Do Not Become Early Warnings: A Negative-Result Dynamical-EVT Study for Public Industrial Monitoring Data

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Abstract

Industrial early-warning studies often begin with pointwise anomaly or exceedance scores, while maintenance decisions are made at the level of alarm episodes, failures, vehicles, or repair actions. This paper uses a negative-result dynamical-EVT study on public industrial monitoring data to test whether score-level extremes survive that conversion. The evaluated assets combine causal state construction, train-split robust scoring, threshold sensitivity, extremal-index diagnostics, declustered alarm episodes, and event-level matching where event labels are available. The primary event-level case studies are MetroPT and MetroPT2; SCANIA Component X is treated as a vehicle-level stress test, while Hydraulic Systems and SECOM are condition-monitoring demonstrations with different targets. The strongest empirical result is also the central limitation: the registered compressor diagnostics recovered the single labelled test event in both MetroPT and MetroPT2, but only with 2527 and 849 predicted alarm episodes and precision below 0.12%. A separate threshold-grid alarm analysis produced zero event recall across all registered quantiles. Hydraulic Systems, SCANIA, and SECOM produced $F_1 = 0$ under the conservative high-quantile diagnostic. These results support a narrow conclusion: dynamical-EVT-inspired clustering diagnostics can expose dependence and alarm-conversion failure modes, but on the evaluated public data they did not provide reliable operational early warning. Useful deployment requires identifiable failure-proximal recurrence, sufficient independent events or entities, stable regime definitions, and alarm policies evaluated at the episode level.

1 Introduction

Industrial maintenance alarms are useful only when they map to the decision unit that operators and engineers actually face. A pointwise anomaly score may flag thousands of telemetry samples around one physical failure, while the operator sees one alarm burden, one repair decision, and one opportunity for useful warning. The same difficulty appears outside compressor telemetry: a fleet repair record, a laboratory hydraulic cycle, and a semiconductor yield label are not exchangeable evidence units. Any empirical claim must therefore separate samples, exceedances, clusters, alarm episodes, failures, vehicles, cycles, and production examples.

Dynamical extreme-value theory offers a tempting route from recurrent machine states to early-warning logic. If an observable is large when the trajectory returns near a target set, then recurrence and periodicity can cluster extremes [4, 5]. In industrial monitoring, however, clustered extremes can also arise from operating regimes, preprocessing, smoothing, missingness, persistent shifts, or alarm-conversion rules. Noise can further weaken the connection between deterministic recurrence and observed extremes [2]. The scientific question is therefore not whether an extremal-index diagnostic can be computed. The question is when such a diagnostic translates into useful episode-level warning.

This paper is framed as a negative-result study. It asks four research questions. First, does threshold and regime conditioning stabilise extremal-index diagnostics relative to global thresh-

olding? Second, does declustering reduce duplicate alarm episodes without destroying event recall or warning lead time? Third, does the registered robust extreme-score diagnostic add operational value after episode conversion? Fourth, which mechanisms explain empirical failure: target-region misspecification, regime mixture, too few independent failures, non-extreme fault signatures, label ambiguity, calibration shift, or alarm conversion?

The evaluated workflow combines causal state construction, training-only target-region construction, regime-conditioned thresholds, extremal-index diagnostics, multivariate lag patterns, alarm merging, event matching, baseline provenance, and dataset-specific condition diagnostics. The interpretation is deliberately conservative so that negative and non-estimable outcomes remain visible.

The contribution is narrow. The study shows that dynamical-EVT-inspired diagnostics can reveal dependence, threshold sensitivity, duplicate-alarm behavior, and calibration limits, but the current public-data evidence does not show reliable early warning. MetroPT and MetroPT2 are the primary event-level case studies; SCANIA Component X is an external vehicle-level stress test; Hydraulic Systems and SECOM are supplementary condition-monitoring demonstrations. This separation is essential because the datasets support different estimands. The paper therefore reports what failed, where the evidence is insufficient, and which conditions would be required before making broader predictive-maintenance claims.

2 Related Work

Predictive-maintenance models often begin with supervised classification, reconstruction error, distance-based anomaly detection, or streaming thresholds. These methods are useful engineering baselines, but their common output is a timestamped score rather than an episode-level maintenance decision. EVT-based streaming detectors such as SPOT use peaks-over-threshold modelling to adapt anomaly thresholds in streams [10]; they address threshold calibration, but they do not by themselves resolve operating-regime mixture, duplicate alarms per failure, or the distinction between clustered observations and independent maintenance events.

Classical declustering and extremal-index estimation address dependence among extremes. Ferro and Segers proposed an inference scheme for clusters of extreme values using inter-exceedance times and the extremal index [3]. In dynamical systems, hitting-time statistics connect returns to target neighborhoods with extreme-value laws [4]. Periodic points can generate an extremal index smaller than one, giving a mathematical route from recurrence to clustered extremes [5]. The finite-sample industrial setting is less clean: noise may make deterministic recurrence appear closer to independence, and observed clustering can also arise from smoothing, missingness, persistence, or regime mixture [2]. Recent multivariate work extends dynamical EVT to cross-sectional dependence among vector-valued observables [1], which motivates the lag-pattern diagnostics used here.

The industrial datasets used by the repository are public condition-monitoring resources with different independence units and labels. MetroPT is an air-production-unit telemetry dataset for predictive maintenance [11]. MetroPT2 is a replication-style benchmark released through Zenodo [12]. SCANIA Component X is a fleet-level multivariate time-series dataset with operational histories and repair records for an anonymized component [7]. UCI Hydraulic Systems provides laboratory hydraulic test-rig cycles with component-condition targets [6]. UCI SECOM provides semiconductor process measurements with pass/fail yield labels [8]. These datasets broaden the empirical base, but they do not remove the need for estimand-specific accounting: one long compressor fault interval, one truck repair history, one hydraulic load cycle, and one wafer-level yield example are not interchangeable statistical units. The methodological gap addressed here is the absence of a reproducible layer that connects dynamical extreme diagnostics to alarm episodes, tabular condition diagnostics, sensitivity analyses, and claim constraints.

3 Method

Let $Z_t \in \mathbb{R}^p$ be the causal state vector at sample t . For compressor telemetry, Z_t contains the registered raw pressure, temperature, current, and flow channels used by the real-data diagnostic. For tabular condition-monitoring adapters, Z_t contains the registered feature vector for the cycle, vehicle, or production example. Scaling is fitted on the training partition only: each feature is centered by the training median and divided by the training interquartile range with a positive floor for zero-spread features.

Operating regimes R_t , when used in the sensitivity assets, are inferred without test labels. Regimes are conditioning variables for threshold estimation and are not interpreted as mechanical causes. The registered real-data matrix reported in Table 7 uses a conservative train-split robust score rather than the full dangerous-region construction.

The implementation also provides dangerous-region scoring for separate diagnostics. The failure-proximal prototype variant uses only training failure intervals and records the source interval for every prototype. The unsupervised rare-state variant ignores failure labels and fits a high-distance rare-state set from training data only. These variants are available for diagnostic analysis and ablation coverage, but the registered real-data matrix in this manuscript is the conservative high-quantile score; it should not be read as evidence that both dangerous-region variants have already been optimized on every dataset.

For a target region A , the scalar observable is

$$X_t = -\log\{d(Z_t, A) + \epsilon\},$$

where d is the robust Euclidean distance in scaled coordinates and $\epsilon > 0$ prevents singularities. Larger X_t means closer proximity to the training-defined region. This is an anomaly or recurrence score, not a calibrated probability of failure.

For each regime r , a threshold $u_r(q)$ is estimated from the declared training or calibration partition at quantile q . Exceedances are $I_t(q, r) = \mathbf{1}\{X_t > u_r(q), R_t = r\}$. Runs declustering merges exceedances separated by at most m samples. The runs estimator and the Ferro–Segers inter-exceedance estimator summarize clustering under the frozen q and m settings [3]. The inverse extremal index is discussed only as an asymptotic cluster-size diagnostic under the assumptions of the estimator.

Pointwise exceedances are converted into alarm episodes by the declared merge gap. A test alarm is a true event alarm only if the episode intersects the predeclared warning window for a held-out failure under the frozen matching tolerance. All event recall, event precision, false-alarm burden, duplicate-alarm count, lead time, time under warning, and utility metrics are computed after this episode conversion when event labels exist. This prevents timestamp counts from being treated as independent maintenance evidence.

The score-stratification target for horizon h is $Y_t^{(h)} = \mathbf{1}\{\text{a labelled event begins within } h \text{ after } t\}$. In the current evidence, model outputs are anomaly ranks or recurrence scores. Probability-calibration language is therefore restricted: Brier-style summaries are reported only as diagnostics of score stratification unless a separate calibration partition contains enough positive independent units for an explicit probability map.

The leakage contract is: training features cannot contain held-out failure samples; scaling, regimes, target regions, thresholds, lag selection, merge-gap selection, and calibration cannot use test events; SCANIA vehicles cannot appear in multiple partitions; and feature windows cannot cross maintenance-reset boundaries. The codebase enforces parts of this contract through split-specific scaling, provenance checks, SCANIA vehicle-leakage checks, and paper/package audits; remaining checks are recorded as required leakage-audit extensions rather than treated

as completed evidence. Under the current registered diagnostic the algorithm is:

1. Split entities or intervals into training, validation, calibration, and test partitions.
2. Fit scaling and robust score thresholds on allowed partitions only.
3. Select q, m , and merge gap on validation/calibration partitions only where used.
4. Freeze all parameters and score the held-out test partition causally.
5. Extract exceedance clusters, merge alarm episodes, and match them to labelled events.
6. Report event metrics, uncertainty by independent unit, and non-estimable cases.

The practical complexity for scoring is $O(npK)$, where n is the number of samples or entities, p is the state dimension, and K is the number of stored prototypes or rare-state summaries. Thresholding and runs declustering are linear in n . The streaming-compatible parts are causal feature construction, scaling with frozen parameters, scoring, thresholding, and alarm merging. Prototype construction, model selection, and final event matching are offline evaluation steps.

4 Simulation Study

The simulation layer is designed to falsify overconfident interpretations before industrial data are used. The registered systems include independent extremes, periodic dynamics, noisy periodic dynamics, regime mixtures, missingness perturbations, and degradation-like cyclic trajectories. Each simulation cell records its configuration, random seed, threshold grid, run-length grid, estimator status, and raw outputs sufficient to regenerate the paper assets.

The main estimands are finite-sample bias and root mean squared error for extremal-index estimators, coverage or failure status where intervals are available, and recovery of clustered episodes. The simulation artifacts deliberately preserve estimator failures. A failed fit is treated as a terminal experiment outcome rather than an infrastructure error, because failure regions are part of the statistical evidence.

The current generated recovery table and figure are interpreted in Section 7 as limited simulation diagnostics. They show that recovery error is measured under registered settings, but they do not support a statement that the same estimator is reliable for every industrial fault interval.

5 Datasets

The datasets are assigned roles before interpretation. MetroPT and MetroPT2 are the primary event-level case studies because they contain ordered compressor telemetry and labelled failure periods [11, 12]. They support early-warning, alarm-episode, recurrence, and per-failure questions, but their independent event count is small. SCANIA Component X is an external vehicle-level stress test with irregular fleet histories, repair labels, and censoring [7]. Hydraulic Systems is a supplementary laboratory condition-monitoring demonstration with load-cycle component states [6]. SECOM is a supplementary high-dimensional production-quality demonstration with wafer-level yield labels [8].

This role assignment prevents a common mistake: row counts do not become independent failures. Table 1 is retained as a descriptive split inventory for the synthetic paper-asset cache. Table 2 records real-data source, independence unit, event or target source, and processed row counts. The independent unit column is the primary evidential quantity for empirical interpretation.

The supported estimands differ. MetroPT-style rows address event-level early warning on temporal telemetry splits. SCANIA addresses vehicle-level repair-risk ranking on a published test split. Hydraulic Systems addresses cycle-level component-condition detection. SECOM addresses production-example yield-failure detection. Hydraulic Systems and SECOM are therefore not interpreted as event-level replications of the compressor task. Broader predictive claims

Table 1: Dataset and split summary.

dataset	split	rows	positive_samples	regimes
synthetic_cyclic	test	4000	401	3
synthetic_cyclic	train	12000	0	3
synthetic_cyclic	validation	4000	2000	3

Table 2: Registered real-data characteristics generated by the real-data matrix.

Dataset	Version	Unit	Event source	Rows
hydraulic_systems	UCI dataset 447	load cycle	cycle-wise component condition targets in profile.txt	2205
metropt	Zenodo record 6854240	failure episode / operating day, not timestamp	version-controlled failure metadata YAML	10773588
metropt2	Zenodo record 7766691	failure episode / operating day, not timestamp	dataset-specific timestamped failure metadata in code	7116940
scania_component_x	Researchdata.se dataset 2024-34	vehicle	published labels and time-to-event files	1516819
secom	UCI dataset 179	wafer / production example	pass/fail yield labels in secom_labels.data	1567

require coherent independent units and comparable targets, not only successful data preparation.

6 Experimental Protocol

The evaluation protocol is frozen before final experiment execution. It declares the train, validation, calibration, and test partitions; the exclusion and maintenance-boundary fields; warning horizons; alarm merge gap; matching tolerance; matching method; bootstrap unit; primary metrics; secondary metrics; and event utility costs.

Primary event metrics are event precision, event recall, event F_1 , false-alarm events per operating day, duplicate alarms per failure, median warning lead time, and horizon Brier score. Secondary metrics include pointwise ranking metrics, time under warning, and event utility. Baselines are required to use the same causal feature availability and the same frozen event conversion, so a model is not rewarded for using future information or a more favorable matching rule.

Uncertainty is attached to the independent unit declared by the dataset and protocol. For simulation cells, the unit is the Monte Carlo repetition or registered replicate. For MetroPT-style case studies, the unit is the failure episode or operating interval, not the timestamp. For SCANIA, the relevant unit is the vehicle or repair history defined by the dataset release. For Hydraulic Systems, it is the load cycle with component-condition labels. For SECOM, it is the wafer or production example with a pass/fail yield outcome. Multiplicity is handled by keeping primary metrics and hypotheses predeclared and by reporting dataset-specific estimands.

The registered real-data matrix has three audit layers. The verification layer checks local raw and processed manifests, row counts, schema summaries, and declared independence units. The industrial-result layer writes one conservative diagnostic row per registered real dataset. The evidence-scope layer combines both outputs into a machine-readable summary of datasets, workflows, sampling structures, independent units, diagnostic statuses, and claim boundaries. This protocol is intentionally strict: a broader dataset base strengthens external coverage only when the estimand differences remain visible.

7 Results

7.1 Simulation validity and estimator stability

The first research question asks whether the extremal-index diagnostics are stable enough to interpret. The current simulation artifact is a limited recovery study rather than the full Monte Carlo validation required for a methodological paper. Figure 1 reports runs-estimator RMSE

Table 3: Dataset roles, sampling structures, independent units, verification status, and diagnostic result status.

Dataset	Sampling structure	Unit	Verification	Industrial result
hydraulic_systems	hydraulic test-rig cycles with multi-rate sensor traces	load cycle	verified	succeeded
metropt	ordered compressor telemetry	failure episode / operating day, not timestamp	verified	succeeded
metropt2	ordered compressor telemetry	failure episode / operating day, not timestamp	verified	succeeded
scania_component_x	fleet-level irregular vehicle histories	vehicle	verified	succeeded
secom	semiconductor process measurements per production entity	wafer / production example	verified	succeeded

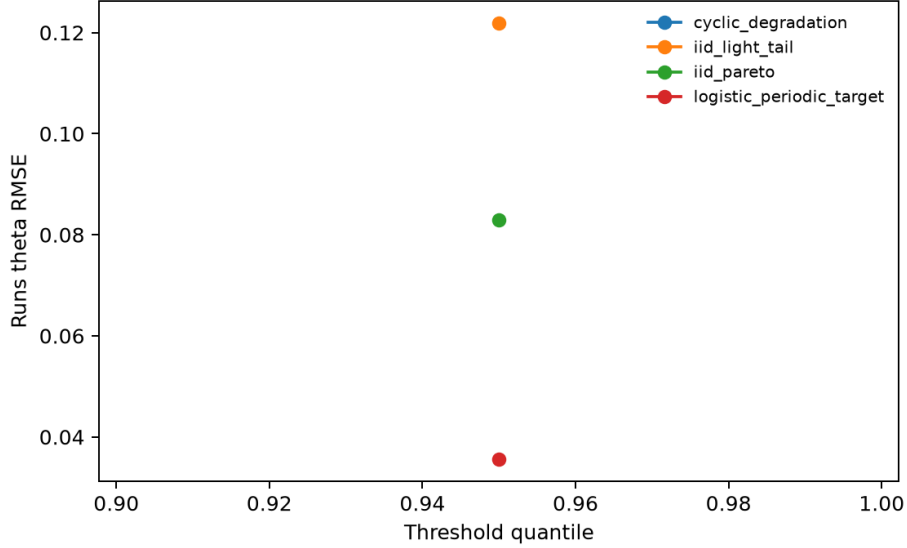


Figure 1: Limited simulation recovery summary. The plotted quantity is runs-estimator RMSE by threshold quantile and simulation family; broader Monte Carlo bias, coverage, and failure-rate validation remains a required extension.

by threshold quantile and simulation family. The result is useful for finding unstable threshold regions, but it is not sufficient to claim estimator reliability across industrial fault mechanisms.

Threshold and run-length sensitivity in Figure 2 and Table 4 show that clustering conclusions depend on q and m . That supports a negative-result interpretation: regime-conditioned dynamical-EVT diagnostics are sensitive design objects, not model-free evidence of impending failure.

7.2 Alarm conversion and failed-detection regions

The second research question asks whether declustering and alarm merging reduce duplicate alarms while preserving event recall. In the registered threshold grid, Figure 3 and Table 5 show zero event recall at all reported quantiles, with false-alarm burdens between 133.92 and 168.48 alarm episodes per operating day. This is a failed-detection region under the current observable, threshold grid, and matching rule; it shows that alarm conversion can destroy utility.

Matching sensitivity in Figure 4 and Table 6 shows whether event F_1 changes under alternative merge gaps and post-onset tolerances. The scientific result is the dependence of conclusions on episode-conversion policy. A setting that looks better only because the matching rule is relaxed is not evidence of earlier warning.

7.3 Real-data diagnostic outcomes

The third and fourth research questions ask whether the registered robust extreme-score diagnostic adds operational value after episode conversion, and why empirical failures occur. Table 7

Table 4: Threshold and run-length stability diagnostics.

threshold_quantile	run_length	n_exceedances	n_clusters
0.9	0	2000	90
0.9	2	2000	41
0.9	5	2000	39
0.9	10	2000	38
0.9	20	2000	38
0.95	0	1000	105
0.95	2	1000	43
0.95	5	1000	34
0.95	10	1000	33
0.95	20	1000	33
0.97	0	600	141
0.97	2	600	44
0.97	5	600	31
0.97	10	600	31
0.97	20	600	31
0.98	0	400	157
0.98	2	400	48
0.98	5	400	34
0.98	10	400	31
0.98	20	400	31
0.99	0	200	128
0.99	2	200	68
0.99	5	200	32
0.99	10	200	31

Table 5: Failed-detection grid for threshold, recall, precision, and false-alarm burden.

threshold_quantile	threshold	event_recall	event_precision	false_alarm_events_per_day	median_warning
0.9	1.119	0	0	168.5	
0.95	1.329	0	0	146.9	
0.97	1.388	0	0	133.9	
0.98	1.411	0	0	146.9	
0.99	1.439	0	0	138.2	

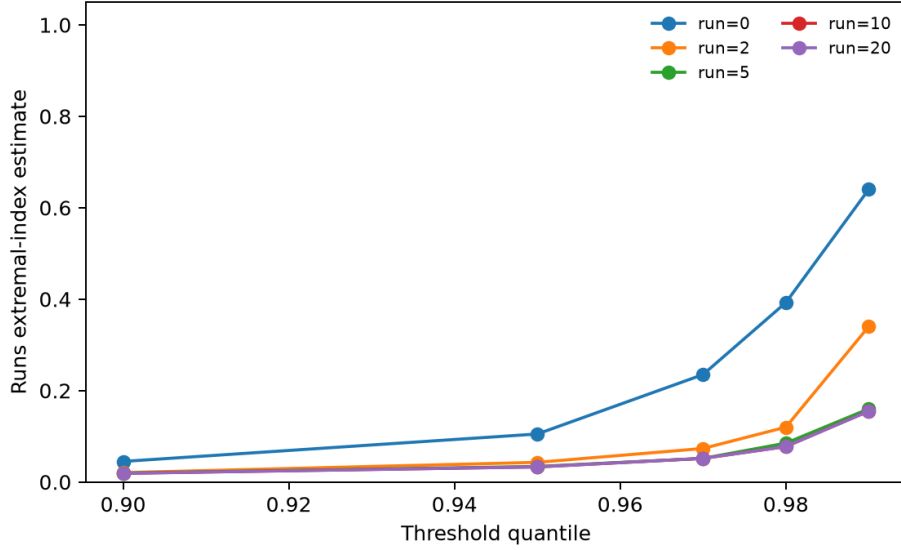


Figure 2: Threshold and run-length stability diagnostic for the registered recurrence observable.

gives the registered diagnostic row for each dataset role. MetroPT and MetroPT2 each contain one labelled test event in the registered split. Both events are detected, but the detection is operationally weak: MetroPT produces 2527 predicted alarm episodes with precision 0.00040, and MetroPT2 produces 849 predicted alarm episodes with precision 0.00118. The median lead times, 3346 and 3469 samples respectively, are not persuasive by themselves because they are paired with extreme false-alarm burdens.

Hydraulic Systems, SCANIA Component X, and SECOM all have $F_1 = 0$ under the conservative high-quantile diagnostic. These are not hidden failures. They are evidence that a simple extreme-score adapter does not transfer across laboratory cycle states, vehicle-level repair risk, and wafer-level yield labels without changing the estimand and evaluation design.

The dominant root-cause classifications are therefore provisional but informative. For MetroPT and MetroPT2, the current evidence points to alarm-conversion failure and insufficient independent failures: recall can be obtained, but only by issuing many alarms around sparse labelled events. For SCANIA, the dominant issue is estimand mismatch: a vehicle-level repair-risk task is not the same as compressor event warning. For Hydraulic Systems and SECOM, the current high-quantile score finds no useful positive-event detection, consistent with non-event labels and signatures that need different feature families or supervised baselines.

7.4 Score stratification and calibration limits

Figure 5 and Figure 6 are retained as score-stratification diagnostics, not probability calibration evidence. The current model outputs are ranks or anomaly scores. Table 8 reports Brier-style summaries by split, but the independent positive-event evidence is too limited for a strong calibrated probability claim.

7.5 Ablations and audit timeline

The ablation summary in Table 9 records coverage of registered sensitivity families. It does not establish that every alternative has been fairly optimized. The event timeline in Figure 7 is retained because aligned raw score, threshold, alarm, and failure displays expose timing mistakes that aggregate metrics hide.

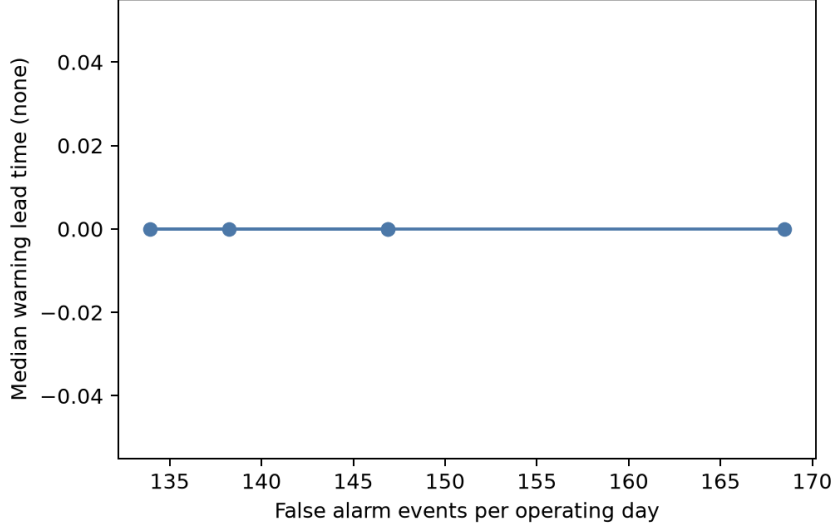


Figure 3: Failed-detection region for the registered alarm policy. The plotted grid has zero event recall, so the figure must not be read as a useful lead-time trade-off.

8 Discussion

The main scientific lesson is negative. Clustered extremes and low extremal-index diagnostics can be observable in industrial telemetry, but that does not imply useful early warning. The conversion from pointwise exceedances to alarm episodes is a separate modelling decision. In the registered threshold grid, that conversion produced a failed-detection region; in the compressor diagnostics, event recall was possible only with thousands of predicted alarm episodes and very low precision.

The likely mechanisms differ by dataset role. For MetroPT and MetroPT2, the limiting factors are sparse independent failures, target-region uncertainty, and alarm burden. A single detected held-out event cannot support a stable estimate of event-level performance when the alarm policy generates many episodes. For SCANIA Component X, the issue is not merely performance; it is a different estimand, because vehicle-level repair-risk ranking is not compressor early warning. Hydraulic Systems and SECOM show another boundary: condition-state and yield labels can be processed by the software, but they do not supply event-onset evidence for dynamical-EVT alarm claims.

The extremal index should therefore be read as a dependence diagnostic. It can identify clustering under a chosen observable, threshold, and run length, but it is not a mechanical degradation variable. Regime mixture, smoothing, missingness, periodic control behavior, and persistent shifts can all create clustered exceedances. Useful deployment would require showing that the target region is identifiable from training data, that failures recur through that region before onset, that enough independent failures or entities exist for uncertainty estimation, and that the alarm policy remains useful after episode conversion.

Calibration is similarly limited. The current outputs are ranks or anomaly scores. Reliability-style plots are useful for detecting score-orientation mistakes and split drift, but they are not evidence of calibrated failure probabilities unless a separate calibration partition contains enough independent positive units and the score-to-probability map is tested on held-out data.

The reproducibility layer remains important because it prevents negative results from being silently edited away. The repository checks generated references, citations, unsupported claim language, compiled PDFs, and artifact manifests. These checks do not make the method scientifically successful. They make the boundary between supported evidence and unsupported

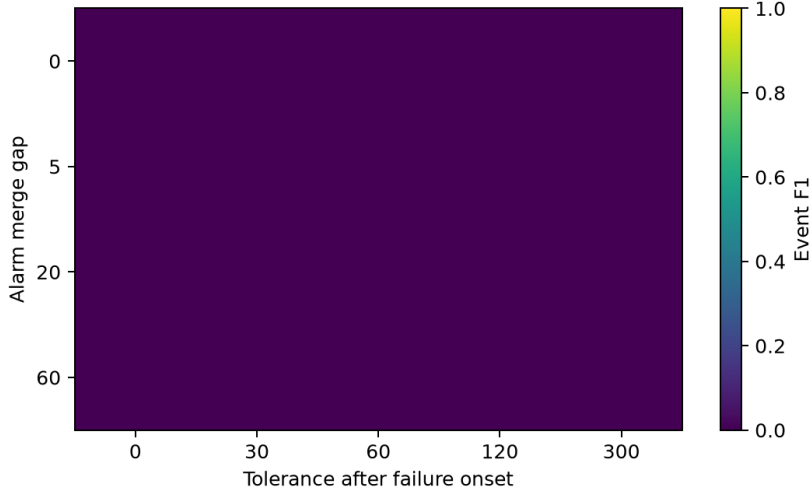


Figure 4: Event F_1 sensitivity over alarm merge gap and post-onset matching tolerance.

interpretation explicit.

9 Conclusion

Dynamical-EVT-inspired clustering diagnostics can reveal dependence structure, threshold sensitivity, and alarm-conversion failure modes, but on the evaluated public industrial datasets they did not reliably translate into operational early warning. The supported contribution is narrow: useful deployment requires identifiable failure-proximal recurrence, sufficient independent events or entities, stable regime definitions, leakage-free tuning, and episode-level alarm policies. The current evidence supports those requirements as boundary conditions, not a claim of broad predictive superiority.

Code and Data Availability

Code, configuration files, data-acquisition scripts, generated-paper tooling, and reproducibility materials are maintained in the project repository [9]. Raw third-party datasets are not redistributed by this repository; the acquisition pipeline stores them locally under `data/raw/` and documents their public sources through the dataset registry.

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Table 6: Event matching tolerance and alarm merge-gap sensitivity.

merge_gap	matching_tolerance_after	event_recall	event_precision	event_f1	false_alarm_events	duplica
0	0	0	0	0	157	0
0	30	0	0	0	157	0
0	60	0	0	0	157	0
0	120	0	0	0	157	0
0	300	0	0	0	157	0
5	0	0	0	0	34	0
5	30	0	0	0	34	0
5	60	0	0	0	34	0
5	120	0	0	0	34	0
5	300	0	0	0	34	0
20	0	0	0	0	31	0
20	30	0	0	0	31	0
20	60	0	0	0	31	0
20	120	0	0	0	31	0
20	300	0	0	0	31	0
60	0	0	0	0	31	0
60	30	0	0	0	31	0
60	60	0	0	0	31	0
60	120	0	0	0	31	0
60	300	0	0	0	31	0

Table 7: Conservative real-data diagnostic rows by dataset role and independent unit.

Dataset	Status	Estimand	Units	Targets	Predictions	F1
hydraulic_systems	succeeded	cycle-level hydraulic component degradation on chronological test split	441	431	0	0.000
metropt	succeeded	event-level early warning on temporal test split	1	1	2527	0.001
metropt2	succeeded	event-level early warning on temporal test split	1	1	849	0.002
scania_component_x	succeeded	vehicle-level repair risk on published test split	5045	142	132	0.000
secom	succeeded	wafer-level semiconductor yield-failure detection on chronological test split	314	17	10	0.000

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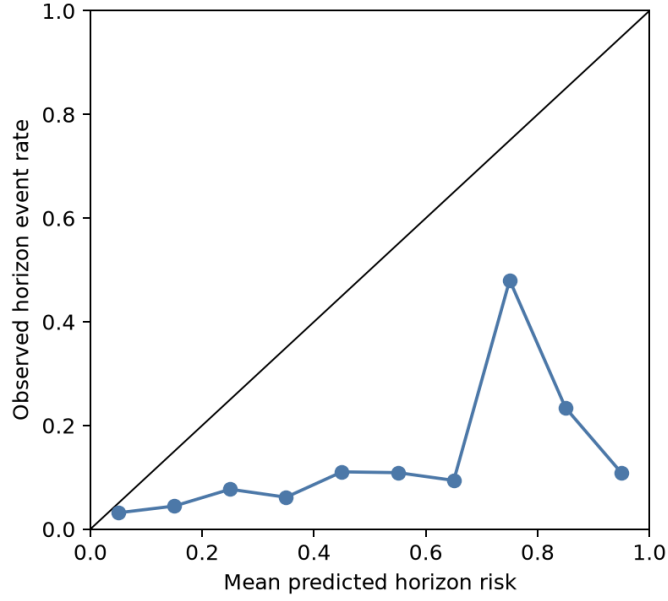


Figure 5: Score-stratification diagnostic for ranked horizon scores. Probability interpretation is not claimed.

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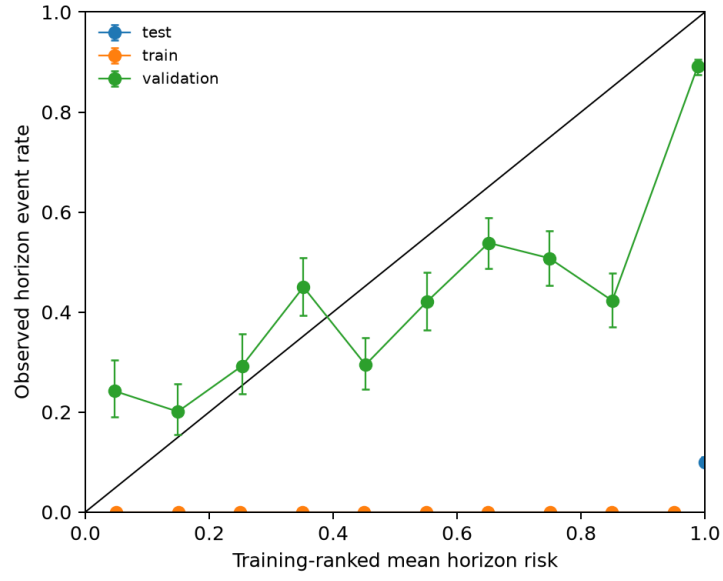


Figure 6: Split-aware score-stratification intervals for ranked horizon scores.

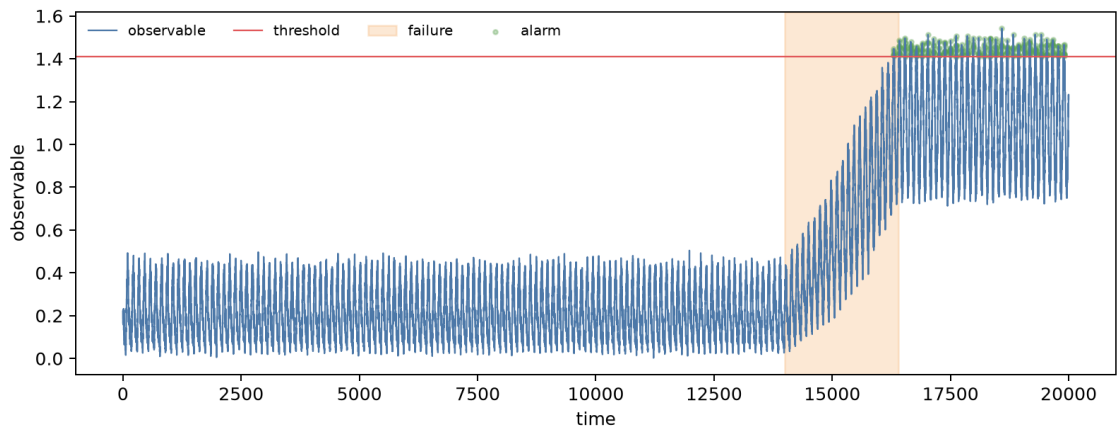


Figure 7: Aligned alarm timeline with failures and threshold exceedances.

Table 8: Split-aware calibration intervals using training-ranked risk scores.

split	lower	upper	count	mean_probability	event_rate	event_rate_ci_lower	event_rate_ci_upper
test	0.9	1	4000	1	0.1003	0.09132	0.1099
train	0	0.1	1199	0.05	0	0	0.003194
train	0.1	0.2	1200	0.15	0	0	0.003191
train	0.2	0.3	1201	0.25	0	0	0.003188
train	0.3	0.4	1199	0.35	0	0	0.003194
train	0.4	0.5	1200	0.45	0	0	0.003191
train	0.5	0.6	1201	0.55	0	0	0.003188
train	0.6	0.7	1200	0.65	0	0	0.003191
train	0.7	0.8	1199	0.75	0	0	0.003194
train	0.8	0.9	1200	0.85	0	0	0.003191
train	0.9	1	1201	0.95	0	0	0.003188
validation	0	0.1	215	0.04765	0.2419	0.1895	0.3033
validation	0.1	0.2	234	0.1494	0.2009	0.1545	0.2568
validation	0.2	0.3	219	0.2529	0.2922	0.236	0.3556
validation	0.3	0.4	280	0.352	0.45	0.3928	0.5086
validation	0.4	0.5	292	0.4522	0.2945	0.2452	0.3492
validation	0.5	0.6	283	0.5512	0.4205	0.3644	0.4787
validation	0.6	0.7	366	0.651	0.5383	0.487	0.5887
validation	0.7	0.8	323	0.7488	0.5077	0.4534	0.5618
validation	0.8	0.9	317	0.8504	0.4227	0.3696	0.4777
validation	0.9	1	1471	0.9883	0.8912	0.8743	0.9061

Table 9: Ablation matrix summary.

ablation_family	setting	summary_value
threshold	0.9	0.1
threshold	0.95	0.05
threshold	0.97	0.03
threshold	0.98	0.02
threshold	0.99	0.01
run_length	0	157
run_length	2	48
run_length	5	34
run_length	10	31
run_length	20	31
lag	0	
lag	1	
lag	2	
lag	5	
lag	10	
regime	off	
regime	loaded	
regime	recovery	
dangerous_region	prototype	
dangerous_region	constraint	
dangerous_region	density	
noise_missingness	none	
noise_missingness	noise_0.05	
noise_missingness	missing_0.05	